

High Reversible Lithium Storage Capacity and Structural Changes of Fe_2O_3 Nanoparticles Confined inside Carbon Nanotubes

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The structural evolution of electrochemically prelithiated Fe_2O_3 nanoparticles confined in carbon nanotubes (CNTs) during lithium insertion/extraction is studied by in situ transmission electron microscopy. It is found that the aggregation and coarsening of Fe core-containing Li_2O ($\text{Fe}@\text{Li}_2\text{O}$) nanograins formed during the charge process are prevented by the spatial restriction of the CNTs. A high reversible capacity of 2071 mA h g^{-1} for the encapsulated Fe_2O_3 nanoparticles in CNTs is demonstrated when the material is used as the anode of lithium ion batteries. This is the highest reversible capacity ever reported for an Fe_2O_3 electrode. The significantly improved lithium storage capacity of the Fe_2O_3 nanoparticles is attributed to the extra lithium storage due to the enhanced interfacial lithium storage and reversible reaction of LiOH to form LiH and solid-electrolyte-interphase conversion originating from the nanoconfinement of CNTs as well as the very small particle size of the $\text{Fe}@\text{Li}_2\text{O}$ nanograins and their good electrical contact with CNTs.

capacity of 1007 mA h g^{-1} has attracted particular attention because of its low cost, abundant availability, and environmental benignity.^[12–16] However, an Fe_2O_3 anode shows rapid capacity fade during cycling due to its intrinsically low electronic conductivity, aggregation, and loss of electrical contact with current collector caused by large volume changes and pulverization occurring during lithium insertion/extraction.^[17] Various approaches have been developed to overcome these shortcomings, and one of the most effective routes is to fabricate an Fe_2O_3 /carbon nanotube (CNT) anode, which shows improved cycling performance and a high rate capability.^[18–26] Although it has been proposed that CNTs may improve the electrical conductivity of the anode and

restrict the large volume change of Fe_2O_3 during charge/discharge,^[27] there has been no direct evidence clearly showing the working mechanism of CNTs in such electrodes.

Here, we prepared two materials: (a) Fe_2O_3 nanoparticles encapsulated in the hollow core of CNTs and (b) Fe_2O_3 nanoparticles anchored on the outer surface of CNTs. We then investigated the structural changes of the Fe_2O_3 nanoparticles inside/outside the CNTs during charge/discharge by in situ transmission electron microscopy (TEM). The unique spatial restriction of CNTs preventing Fe core-containing Li_2O ($\text{Fe}@\text{Li}_2\text{O}$) nanograins within the CNTs from aggregation and growth during the charging process was directly observed. A high reversible capacity of $2071.3 \text{ mA h g}^{-1}$ for the encapsulated Fe_2O_3 nanoparticles was demonstrated when serving as the anode of LIBs, which is the highest reversible capacity ever reported for Fe_2O_3 electrode materials.

1. Introduction

Rechargeable lithium-ion batteries (LIBs) have aroused extensive research due to their wide-range use or possible use as the power source of portable electronic devices, electric vehicles, and hybrid electric vehicles.^[1–6] Although LIBs have the highest energy density available from existing commercial rechargeable battery systems, their performance still lags behind the increasing demands of newly emerging applications. To meet these demands, it is essential to explore novel electrode materials with high energy and power densities. Recently, transition metal oxides have been found to show desirable lithium storage properties based on a conversion reaction mechanism. Insertion/extraction in excess of 6 Li^+ per formula unit of active material can be achieved, which makes metal oxides a promising candidate anode for high performance LIBs.^[7–11] Among them, Fe_2O_3 with a theoretical

2. Results and Discussion

CNTs with open ends and a uniform diameter of $\approx 20 \text{ nm}$ were prepared by an anodic aluminum oxide (AAO) template method.^[28] Fe_2O_3 nanoparticles were then filled into the hollow core of the CNTs using a simple wet chemistry technique.^[26] As a result, Fe_2O_3 nanoparticle-filled CNTs were obtained (see the Experimental Section). Typical TEM images of the sample are shown in Figure 1a and Figure S1a in the Supporting Information. We can see that Fe_2O_3 particles with

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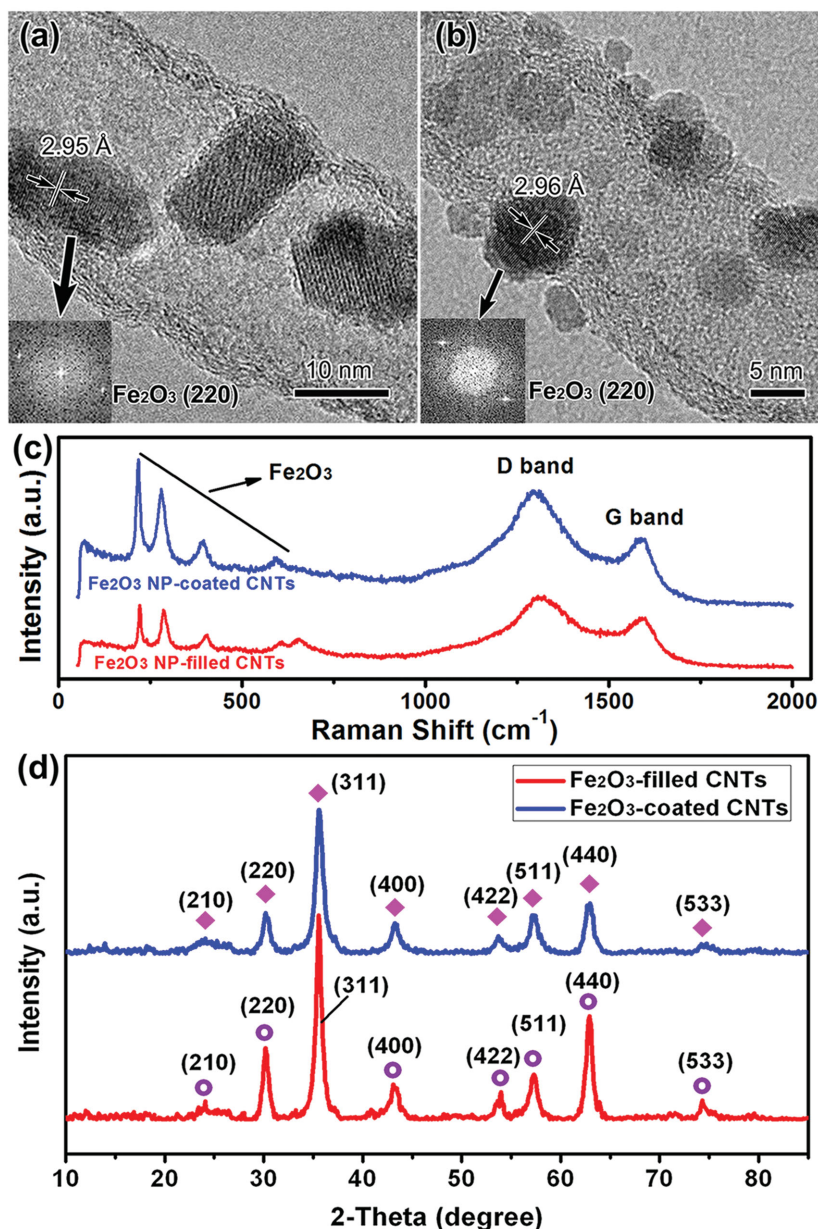


Figure 1. a,b) TEM images, c) Raman spectra, and d) XRD patterns of the Fe_2O_3 nanoparticle-filled and Fe_2O_3 nanoparticle-coated CNTs. The insets in (a) and (b) are corresponding FFT patterns, showing that the Fe_2O_3 nanoparticles are crystallized.

a mean diameter of ≈ 10 nm homogeneously fill the hollow core of the CNTs. For comparison, we also prepared an Fe_2O_3 nanoparticle-coated CNT sample, where Fe_2O_3 nanoparticles were uniformly dispersed on the outer surface of the CNTs (Figure 1b and Figure S1b, Supporting Information). High-resolution TEM (HRTEM) images and their fast Fourier transform (FFT) patterns (insets of Figure 1a,b) revealed an interplanar spacing of 2.95 ± 0.01 Å, corresponding to the (220) lattice plane of the maghemite-C ($\gamma\text{-Fe}_2\text{O}_3$) phase for these two samples, which are in good agreement with their selected area electron diffraction (SAED) patterns (Figure S1c,d, Supporting Information). Energy-dispersive X-ray spectroscopy analysis (Figure S2,

Supporting Information) confirms that both Fe_2O_3 nanoparticle-filled CNTs and Fe_2O_3 nanoparticle-coated CNTs consist of C, O, and Fe. The atomic ratios of the Fe and O in these two samples are 0.38 and 0.71 (Table S1, Supporting Information), respectively. The lower atomic ratio of the Fe and O in Fe_2O_3 nanoparticle-filled CNTs is ascribed to the excess oxygen from the functional groups on the surface of CNTs introduced during the removal of AAO template.^[29] The Raman spectra of Fe_2O_3 nanoparticle-filled and coated CNTs (Figure 1c) show some peaks in the range of $200\text{--}700\text{ cm}^{-1}$, which can be assigned to the Fe_2O_3 . Intensity ratios of D band to G band (I_D/I_G) of 1.38 and 1.55 were demonstrated for these two samples, suggesting a low degree of crystallization of the CNTs.^[27] Nevertheless, the defects may facilitate the insertion and transfer of Li ions in CNTs when the hybrid was used as an electrode. X-ray diffraction (XRD) patterns of the Fe_2O_3 -filled and Fe_2O_3 -coated CNTs are shown in Figure 1d. Diffraction peaks corresponding to the maghemite-C (JCPDS card No. 39-1346) phase are identified for the Fe_2O_3 -filled and Fe_2O_3 -coated CNT samples, consistent with the TEM observations and SAED patterns. No distinct diffraction peaks from $\alpha\text{-Fe}_2\text{O}_3$ (hematite) were observed. This is because $\gamma\text{-Fe}_2\text{O}_3$ is much more stable than $\alpha\text{-Fe}_2\text{O}_3$ for nanoparticles with a size of ≈ 10 nm due to their different Gibbs free energies derived from the excess surface contributions of these two phases.^[30,31] It is also worth noting that no diffraction peak corresponding to graphitic carbon was observed, and it is in good agreement with the result of Raman measurement. This is because the walls of the CNTs synthesized by the AAO template method are poorly crystallized. These carbon layers may facilitate the access of lithium ions to the Fe_2O_3 nanoparticles.

Recently, in situ TEM studies on electrochemical reactions have been conducted to investigate the structural changes in nanostructured electrode materials during the lithiation/delithiation process.^[32–37] CNTs can also act as nano-reactors due to their well-defined 1D tubular structure and good structural stability.^[29,38] We therefore constructed a nanobattery using the prelithiated Fe_2O_3 -filled CNTs as working electrodes to show the confinement effect of CNTs and the structural changes of prelithiated Fe_2O_3 nanoparticles during charge/discharge. Because it is difficult to capture and record structural changes of nanometer-sized electrode materials in the HRTEM when a normal liquid phase electrolyte is used, an all-solid-state Li ion nanodevice using a solid Li/Li₂O compound as the lithium source and a solid-state electrolyte has been developed. It permits high spatial resolution observations of the electrode

structure during lithiation and delithiation.^[39–43] However, the introduction of metallic lithium as an electrode material for the in situ electrochemical reaction makes the lithiation/delithiation process quite different from that of real lithium-ion batteries.^[39] Lithium-ion rechargeable batteries have been termed as “rocking-chair batteries,” and can be described as concentration cells where the lithium ions “rock” from one side to the other under a driving force derived from the electric potential.^[1,44] Here, we constructed an all-solid-state CNT-based lithium-ion concentration cell in a TEM that had no lithium-metal counter electrode by using an electrochemically prelithiated Fe_2O_3 -filled CNT as the working electrode in which enough Li is provided for the following lithiation and delithiation processes (see the Experimental Section). An extra electric potential was applied to drive the insertion and extraction of lithium ions in the electrode. A schematic showing the construction of such a nanobattery and a corresponding TEM image are shown in Figure 2. The prelithiated Fe_2O_3 -filled CNTs were attached to a fixed biased gold wire electrode of a TEM-scanning tunneling microscope (STM) holder. No extra metallic lithium is needed for the lithium source in this case. A protruding CNT was selected and attached as a bridge to a movable gold tip electrode driven by a piezoelectric tube. Bias voltages of $+(4\text{--}5)$ V were applied to the CNT upon charging and $-(4\text{--}5)$ V upon discharging.

The Fe_2O_3 -filled CNTs were electrochemically prelithiated before inserting them into the TEM chamber (see the Experimental Section). Electrochemical full lithiation reaction of the Fe_2O_3 nanoparticles in CNTs mainly consists of three stages: (1) lithium insertion in Fe_2O_3 , (2) lithium induced reduction reaction of Fe_2O_3 forming nano-Fe and Li_2O , and (3) excess lithium storage after conversion reaction. The main structural difference between the prelithiated Fe_2O_3 and fully lithiated Fe_2O_3 is that the nano-Fe uniformly dispersed in Li_2O matrix in the former tends to aggregate, and the nano-Fe and Li_2O grains turn to be larger and well-crystallized after full lithiation. The prelithiated Fe_2O_3 -filled CNTs were not thoroughly lithiated because of the electrochemical nonequilibrium lithiation reaction. No obvious

lattice fringes were observed in the lithiated Fe_2O_3 particles after prelithiation, possibly due to the occurrence of a reaction with Li to form Li_2O and uniformly-dispersed nanoscale Fe (Figure 3a and Figures S3 and S4a, Supporting Information). In the in situ TEM charging process, a positive bias voltage was applied to the CNT. Under the driving force of this external potential, lithium ions migrate from the positive potential end of the CNT along the carbon wall to the amorphous lithiated phase (red arrows in Figure 3b). Meanwhile, electrons are transported in the opposite direction along the wall of the CNT (blue arrow in Figure 3b). As can be seen in Figure 3b,c, the volume of the electrochemically prelithiated Fe_2O_3 compound starts to increase (see black arrows) because more Li^+ was inserted into the compound when meeting electrons from the low potential end. Nanoclusters of $\text{Fe@Li}_2\text{O}$ grains with sizes of 1–2 nm were formed during the accumulation of Li and the aggregation of dispersed nano-Fe (Figure 3d and Figure S4b and Movie S1, Supporting Information), with a total volume expansion of $\approx 52\%$ upon full lithiation. The $\text{Fe@Li}_2\text{O}$ nanocluster was confined in the core of the CNT and made good contact with it. It is worth noting that extra lithium storage may occur by a heterogeneous interfacial reaction due to charge separation at the $\text{Fe/Li}_2\text{O}$ phase boundary (Figure S5, Supporting Information).^[45] Other mechanisms may be involved, such as a reversible solid-electrolyte-interphase (SEI) formation/decomposition and reversible reaction of generated LiOH with Li to form Li_2O and LiH .^[46] The inner space of the CNTs accommodates the volume expansion of prelithiated Fe_2O_3 nanoparticles during the charging process and guarantees desirable cyclic performance of the anode. By reversing the applied bias, discharge of the electrode was observed, accompanied by volume contraction, disappearance of $\text{Fe@Li}_2\text{O}$ nanoclusters (Figures S4c and S6 and Movie S2, Supporting Information), and the oxidation of Fe to Fe^{3+} .

Figure 4 shows typical TEM images of the prelithiated Fe_2O_3 nanoparticles at different charge/discharge stages and the corresponding SAED patterns. We can see that initially, homogeneously dispersed nano-Fe exists in the Li_2O matrix

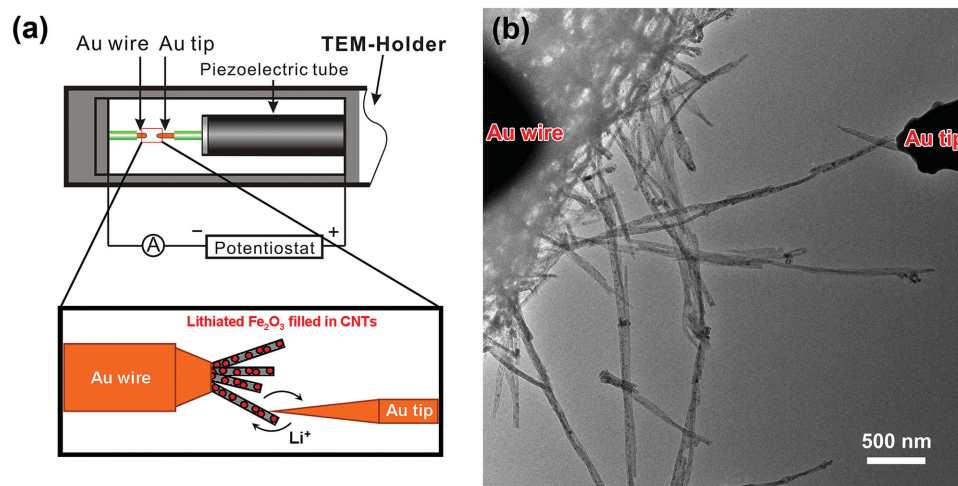


Figure 2. a) Schematic showing the construction of a nanobattery inside a TEM. b) Low magnification TEM image of a Au tip, a Au wire, and bridging prelithiated Fe_2O_3 nanoparticle-filled CNTs.

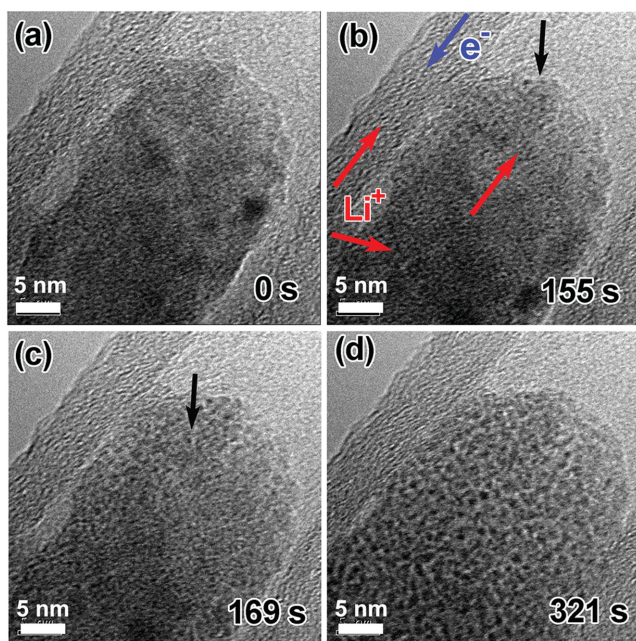


Figure 3. Series snapshots of a prelithiated Fe_2O_3 nanoparticle in a CNT during the Li^+ insertion process upon charging, showing obvious volume swelling and the emergence of a $\text{Fe@Li}_2\text{O}$ nanocluster consisted of nanograins. a) The prelithiated Fe_2O_3 nanoparticle confined inside a CNT. b) and c) The Li^+ insertion of the prelithiated Fe_2O_3 nanoparticle during charging. d) The prelithiated Fe_2O_3 nanoparticle after full charging.

formed after electrochemical prelithiation of the Fe_2O_3 nanoparticles (Figure 4a). Figure 4e shows the corresponding SAED pattern of the prelithiated Fe_2O_3 inside the CNT. The d-spacings of 2.67 and 1.64 Å determined from the pattern are consistent with the lattice spacings of the (111) and (220) planes of Li_2O (JCPDS File No. 77-2144). The lattice spacings for the obscure diffuse rings in the SAED pattern were calculated to be 2.04 and 1.41 Å, which are in good agreement with the spacings of the (110) and (200) planes of body-centered cubic (bcc) Fe (JCPDS File No. 87-0722).^[47] During the first charge, $\text{Fe@Li}_2\text{O}$ nanoclusters were formed accompanied by a volume expansion due to further Li insertion and aggregation of nano-Fe (Figure 4b). The d-spacing calculated from the diffuse rings in the pattern of Figure 4f does not change as a result of the extra Li insertion. But their intensity becomes much stronger, implying that the bcc nano-Fe and Li_2O grains in the $\text{Fe@Li}_2\text{O}$ nanoclusters grow larger and become more crystalline. After Li^+ extraction in the discharge process, as shown in Figure 4c, the $\text{Fe@Li}_2\text{O}$ nanoclusters completely vanish along with the electrochemical oxidation of Fe to Fe_2O_3 . The diffraction rings in Figure 4g can be well indexed as the rhombohedral phase of Fe_2O_3 (JCPDS File No. 01-1053), confirming good reversibility of the Fe_2O_3 conversion reaction. It is worth noting that rhombohedral Fe_2O_3 nanoparticles with sizes up to ≈ 6 nm stably existed after the delithiation (Figure S7, Supporting Information). This is possibly because that more Gibbs free energy can be offered to the surface of as-formed rhombohedral Fe_2O_3 when discharged inside the high potential environment of the inner cavity of

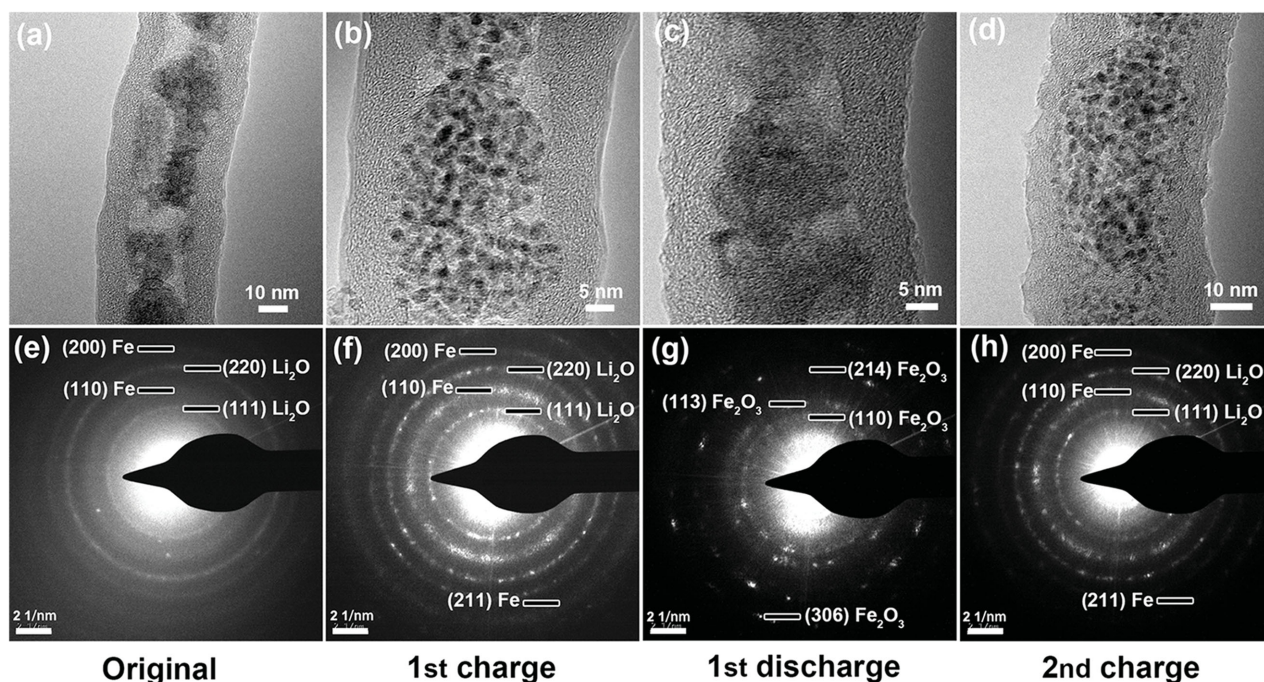


Figure 4. TEM images of a) the original prelithiated Fe_2O_3 nanoparticles confined in a CNT, and their structural change after b) the 1st charge, c) the 1st discharge, and d) the 2nd charge. e–h) The corresponding SAED patterns.

CNTs. As a result, the rhombohedral phase Fe_2O_3 was thermodynamically more favorable and was retained as the main product after delithiation. Normally, the thermodynamically irreversible extraction of Li from Li_2O would cause a capacity loss during the transformation of Fe^0 to Fe^{3+} ,^[12] but in this case it does not happen. A CNT can be regarded as a tubular structure composed of rolled-up graphene layers. Deviation from planarity causes partial rehybridization of the C—C bonds from sp^2 to sp^3 and the pyramidalization of carbon atoms in curved aromatic systems.^[48,49] The π electron density is shifted from the concave inner to the convex outer surface, causing an electron-deficient concave surface of the nanotube.^[48–50] Therefore, when the nanoclusters of $\text{Fe@Li}_2\text{O}$ grains originating from the Li insertion of prelithiated Fe_2O_3 are confined inside the CNT, the interaction between the electron-deficient concave inner surface of the CNT and the anionic oxygen in Li_2O could lead to weakened bonding strength of Li_2O itself and hence lower the activation energy for its decomposition and electrochemical reduction during delithiation,^[50] which facilitates the fully oxidative reaction of Fe^0 to Fe^{3+} rather than to Fe^{2+} ^[47] and improves the overall reversible Li storage capacity of Fe_2O_3 as a result of the conversion reaction. In addition, the $\text{Fe@Li}_2\text{O}$ nanoclusters can be well recovered after the second charge process (Figure 4d,h), indicating good reversibility and cyclic stability of Fe_2O_3 as the anode of LIBs when encapsulated in CNTs.

We also constructed a nanobattery using CNTs coated with prelithiated Fe_2O_3 nanoparticles as a working electrode, and its structural variation during lithiation/delithiation is shown in Figure 5 and Movie S3 in the Supporting Information. Figure 5a shows a pristine prelithiated Fe_2O_3 coated CNT, and

we can see that the CNT is wrapped by a thin layer of prelithiated Fe_2O_3 . During the charge process (Figure 5b,c), some dark $\text{Fe@Li}_2\text{O}$ nanograins of diameter 5–6 nm appeared, which are much larger than those formed inside CNTs as shown in Figures 3 and 4. The total volume of the compound was expanded by $\approx 35\%$ after full lithiation. In the discharge process (Figure 5d–f), the formed $\text{Fe@Li}_2\text{O}$ nanograins gradually disappeared upon Li^+ extraction. Finally, an iron oxide compound was formed with a volume contraction of $\approx 31\%$. Note that not all the fully lithiated Fe_2O_3 on the outer surface of CNTs is completely delithiated after discharge (Figure S8, Supporting Information), implying poor cyclic stability of this material.

The electrochemical lithium storage ability of the Fe_2O_3 nanoparticle/CNT materials was evaluated by a galvanostatic discharge/charge technique. The initial two-cycle discharge–charge curves at 50 mA g^{-1} within a cut-off voltage window of 0.01–2.8 V are shown in Figure 6a. In the first cycle, the discharge and charge capacities are evaluated to be 3037 and 1143.8 mA h g^{-1} for the Fe_2O_3 nanoparticle-filled CNTs and 2462.6 and 1092.3 mA h g^{-1} for the Fe_2O_3 nanoparticle-coated CNTs, respectively. Both of these samples showed a low initial Coulombic efficiency. The irreversible capacity loss can be mainly attributed to the irreversible reaction of lithium with oxygen-containing functional groups and the decomposition of electrolyte to form an SEI layer.^[19,27] During the second cycle, the Coulombic efficiency of the two samples increased up to $\approx 80\%$. From the discharge curve, a voltage plateau at $\approx 0.8 \text{ V}$ in the first cycle corresponding to the electrochemical reaction between Fe_2O_3 and lithium disappears and another plateau corresponding to the final reduction of iron oxide to Fe is retained. This means that an irreversible structural

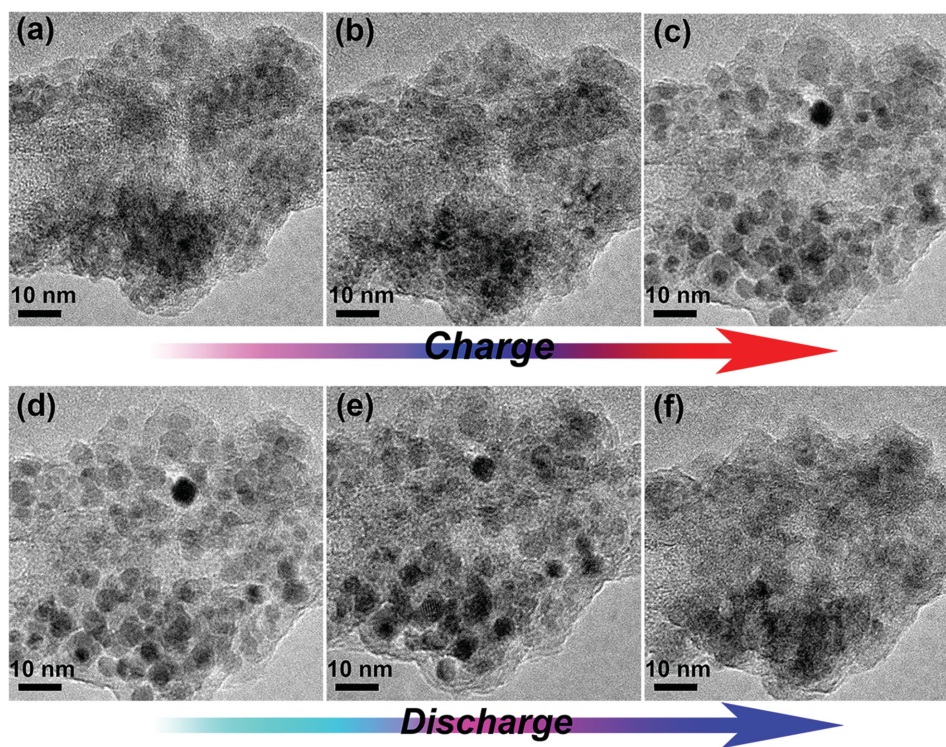


Figure 5. Structural evolution of the CNTs coated with Fe_2O_3 nanoparticles: a–c) lithiation; d–f) delithiation.

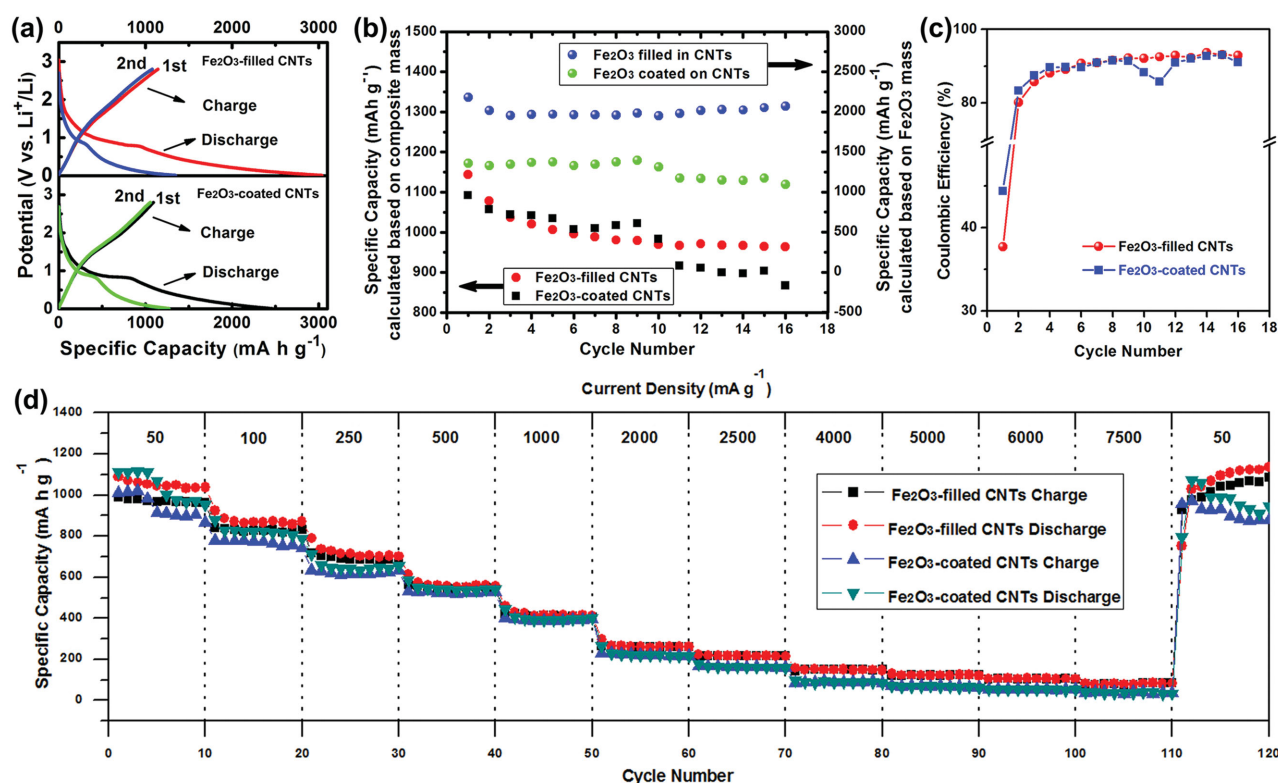


Figure 6. a) Galvanostatic charge/discharge voltage profiles of the Fe₂O₃ nanoparticle-filled and -coated CNTs in the voltage range of 0.01–2.8 V (versus Li⁺/Li) at a current density of 50 mA g⁻¹. b) Reversible lithium storage capacity versus cycle number for the Fe₂O₃ nanoparticle-filled CNTs and Fe₂O₃ nanoparticle-coated CNTs based on the total and Fe₂O₃ mass, at a current density of 50 mA h g⁻¹. c) The Coulombic efficiency versus cycle number for the Fe₂O₃ nanoparticle-filled and -coated CNTs. d) The galvanostatic cycling performance of the Fe₂O₃ nanoparticle-filled and coated CNTs under different current densities.

change for Fe₂O₃ either filled in or coated on CNTs occurred during the first discharge. The reversible charge capacity versus cycle number is shown in Figure 6b. The Fe₂O₃ nanoparticle-coated CNTs with a mass fraction of 45% Fe₂O₃ (Figure S9a, Supporting Information) exhibited an initial specific capacity of 1092.3 mA h g⁻¹ at a current density of 50 mA h g⁻¹ based on the total mass. However, the capacity decreased to 867.2 mA h g⁻¹ after 16 cycles. Meanwhile, a higher reversible specific capacity of 1143.8 mA h g⁻¹ was achieved for the Fe₂O₃-filled CNTs (containing ≈20 wt% Fe₂O₃; Figure S9a, Supporting Information) in the first cycle, and the capacity tended to stabilize at 963.7 mA h g⁻¹ after 16 cycles. This could be attributed to the spatial confinement of the CNTs that prevented the Fe₂O₃ nanoparticles from aggregation, delamination, and loss of electrical contact with the current collector during charge/discharge.^[26] In addition, the specific capacity with respect to the mass of Fe₂O₃ was calculated by subtracting the capacity of pure CNTs shown in Figure S9b in the Supporting Information. The Fe₂O₃ nanoparticle-filled CNTs delivered a high and stable reversible capacity of 2071.3 mA h g⁻¹ based on the mass of Fe₂O₃ after 16 cycles. To the best of our knowledge, this is the highest reversible capacity reported for Fe₂O₃-based anode materials under similar conditions. Unlike Fe₂O₃ nanoparticles confined in CNTs, the Fe₂O₃ located outside the CNTs exhibited a reversible capacity of only 1096.8 mA h g⁻¹, which is in agreement with the theoretical capacity value of Fe₂O₃. These results indicate that

the reversible capacity of Fe₂O₃ can be significantly improved when Fe₂O₃ nanoparticles fill the hollow cores of CNTs. It is worthy of noting that the Coulombic efficiencies of the Fe₂O₃ nanoparticle-filled CNTs (Figure 6c) were higher than ≈93% after the initial few cycles, which is much better than those of the Fe₂O₃ nanoparticle-coated CNTs, indicating a more stable electrode/electrolyte interface of the Fe₂O₃ nanoparticle-filled CNT electrode. The low Coulombic efficiency observed in the first cycle (Figure 6a) could be improved by forming a passivation layer on the electrode surface and introducing closed pores that allow the access of Li⁺ without electrolyte.^[51,52] Moreover, the galvanostatic cycling performance of the Fe₂O₃ nanoparticle-filled and -coated CNTs under different current densities for a total of 120 cycles was studied. As shown in Figure 6d, reversible capacities of 967.2 and 916.5 mA h g⁻¹ (at the fifth cycle) based on the total mass were achieved at a current density of 50 mA g⁻¹ for the Fe₂O₃ nanoparticle-filled and -coated CNTs, respectively. The specific capacities were then decreased upon increasing the current rate of discharge/charge stepwise. It can be seen that both of the Fe₂O₃ nanoparticle-filled and -coated CNTs deliver a stable capacity under each current density, even up to 7500 mA g⁻¹. Importantly, when the current rate was decreased to 50 mA g⁻¹, a reversible capacity of 1084.7 mA h g⁻¹ (at the 120th cycle) for the Fe₂O₃ nanoparticle-filled CNTs was obtained, showing its good electrochemical reversibility and structural stability. The observed slight capacity rise (112% retention of its capacity at

the fifth cycle) is likely to be related to the formation of a gel-like layer or further lithium storage as a result of interface reaction due to the charge separation at the nano-Fe/Li₂O boundary in the process of activation and stabilization of the electrodes.^[19,27] We note that a more stable cyclic performance was achieved for the Fe₂O₃ nanoparticle-filled CNTs in comparison to that of Fe₂O₃ nanoparticle-coated CNTs after the cycling at 7500 mA g⁻¹, indicating a more reliable structure of the former.

The high reversible capacity of Fe₂O₃ confined in CNTs can be mainly attributed to the following factors (Figure 7): first, thorough lithiation/delithiation of the Fe₂O₃ with a highly reversible conversion reaction between Fe³⁺ (Fe₂O₃) and Fe⁰ when confined in the CNTs (as shown in Figure 4); second, extra lithium storage from the enhanced interfacial reaction at the space charge layer;^[45,53–56] and third, reversible conversion reaction of LiOH to form LiH and SEI formation/decomposition.^[46] Theoretically, Fe₂O₃ has a maximum lithium uptake of 6 Li, corresponding to 1007 mA h g⁻¹ specific capacity from the reduction of Fe³⁺ to Fe⁰ based on the conversion reaction. However, the specific capacity of Fe₂O₃ obtained in this study is much higher than this theoretical capacity. This could be due to an extra capacity related to the charge interfacial storage based on the two-phase capacitive behavior of the Li₂O/Fe interface that allows for the storage of Li⁺ on the Li₂O compound side while electrons are localized on the Fe side, thereby formally leading to a charge separation (Figure S5, Supporting Information).^[45,54–56] The total specific capacity of Fe₂O₃ can be calculated from Equations (1) and (2)

$$C_{\text{total}} = C_{\text{conversion}} + C_{\text{interface}} + C_{\text{surface}} \quad (1)$$

$$C_{\text{interface}} = K(S_{\text{interface}}) \quad (2)$$

where C_{total} , $C_{\text{conversion}}$, $C_{\text{interface}}$, and C_{surface} are, respectively, the total specific capacity, conversion reaction capacity, interfacial storage capacity, and surface capacities from reversible reaction of LiOH to form LiH and SEI conversion. $S_{\text{interface}}$ is the total area of the Li₂O/Fe interface in the Fe@Li₂O nanoclusters. $C_{\text{interface}}$ is a function of $S_{\text{interface}}$ and K is the coefficient of the incremental function. The higher the interfacial area, the larger the total reversible capacity per unit mass of Fe₂O₃ in terms of Li storage by the interfacial reaction. The reversible capacity of interfacial lithium storage was improved by the interfacial area that is related to the size of Fe@Li₂O nanoclusters and can be calculated as follows

$$M_{\text{Fe}} = N\rho(4/3)\pi R^3 \quad (3)$$

$$S_{\text{interface}} = N4\pi R^2 \quad (4)$$

where M_{Fe} is the mass of Fe in the Li₂O, N is the number of Fe@Li₂O nanograins in nanoclusters, ρ is the mass density of the Fe core, and R is the radius of the Fe core. The surface area of Fe equals the interfacial area at the Li₂O/Fe interface. The interfacial storage capacity of Fe₂O₃ can be calculated as: $C_{\text{interface}} = K(S_{\text{interface}}) = K[(1/R)(3M_{\text{Fe}}/\rho)]$. Thus, the interfacial Li-storage capacity can be increased by reducing the size of the Fe@Li₂O nanograins. According to the in situ TEM studies, we can see that the prelithiated Fe₂O₃ on the outer surface of CNTs underwent the same structural transformation as that inside the CNTs during the charge/discharge process. However, the Fe@Li₂O nanograins formed outside CNTs have sizes of 5–6 nm, three to six times larger than those formed inside the CNTs. This result clearly shows that the well-defined nanochannels of the CNTs are able to prevent Fe@Li₂O nanograins from growth and aggregation during the charge process, ensuring that the interfacial Li storage reaction can be effectively enhanced.

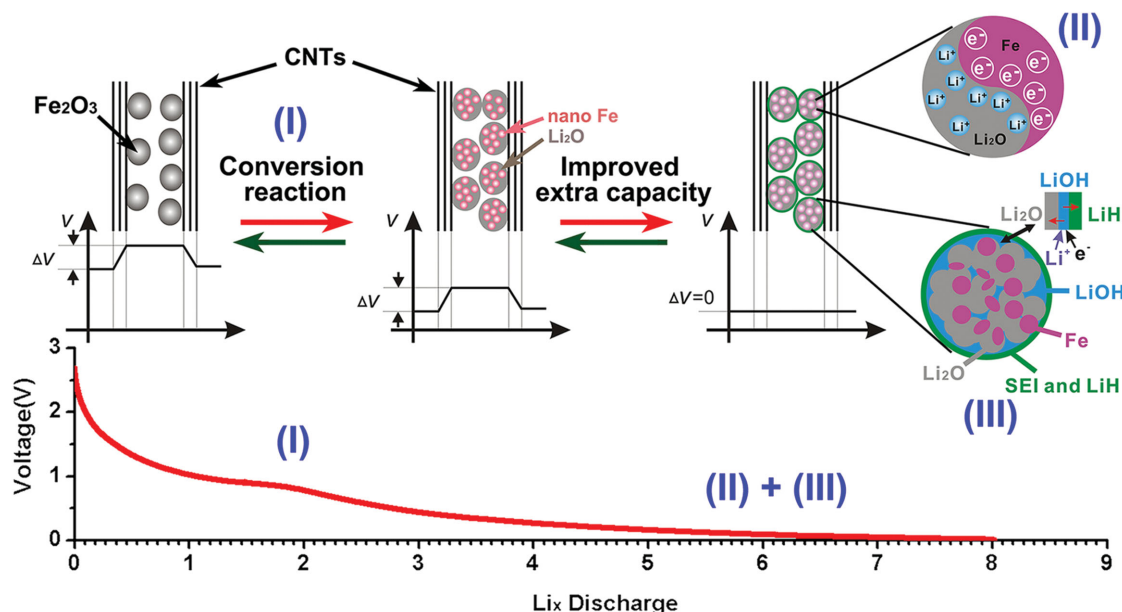


Figure 7. Illustration showing the origin of the improved lithium storage capacity of Fe₂O₃ nanoparticles confined inside CNTs: I, conversion reaction between Fe₂O₃ and nano-Fe⁰; II, improved Li interfacial reaction; III, reversible reaction of LiOH to form LiH and SEI formation/decomposition. V is the electrode potential, and ΔV is the electric potential difference between the concave interior and the convex exterior surface of the CNT.

In addition, the curvature of the CNT wall induces a shift of π electron density from the concave interior to the convex exterior surface, which results in an electric potential difference (see Figure 7).^[48–50,57] The electron-deficient interior surface of the CNTs shows a more positive electric potential than that of electron-enriched exterior surface.^[58,59] Therefore, excess lithium for interfacial storage after the conversion reaction of Fe_2O_3 confined inside CNTs is needed when the electrode potential of the CNT interior approaches the potential value outside CNTs during charging (Figure 7). As a result, more lithium can be accommodated in the boundary region of $\text{Fe}/\text{Li}_2\text{O}$ by this interfacial effect when Fe_2O_3 is confined inside the CNTs. Meanwhile, the interaction between the electron-deficient inner surface of the CNT and as-formed species such as Li_2O and LiH could lead to weakened bonding of Li_2O and LiH , which facilitates their reversible decomposition (Figure 7),^[46] and hence increases the reversible specific capacities.

3. Conclusions

In summary, the structural changes in electrochemically prelithiated Fe_2O_3 nanoparticles confined in and coated on CNTs were studied by in situ TEM. A highly reversible conversion reaction between Fe^0 and Fe^{3+} (Fe_2O_3) during lithiation/delithiation was observed. It was found that the size of the $\text{Fe}@ \text{Li}_2\text{O}$ nanograins formed during the charge process inside the hollow core of the CNTs is much smaller than those for coated on the outer surface. Therefore, the reversible specific capacity of Fe_2O_3 confined in the CNTs can be greatly improved due to the enhanced interfacial lithium storage and reversible reaction of LiOH to form LiH and SEI conversion originating from the nanoconfinement of CNTs. A reversible lithium storage capacity of 2071 mA h g^{-1} was demonstrated for the Fe_2O_3 nanoparticle filled in CNTs as an anode material for LIBs, much higher than the theoretical value. These results provide useful guidance for the design of anode materials for LIBs with a large capacity and good cycling performance.

4. Experimental Section

Chemicals: High-purity aluminum sheets (thickness: $250 \mu\text{m}$; purity: 99.99%) used as starting material for the preparation of CNTs by the AAO method were purchased from Beijing Mengtai Youyan Technology Development Center (Beijing, China). Perchloric acid (70%) was purchased from Shanghai Jin Lu Chemical Co., Ltd. (Shanghai, China). All other chemicals and solvents were purchased from Sinopharm Chemical Reagent Co., Ltd. (Shanghai, China). All the chemicals were analytical grade reagents used without further purification.

Preparation of Fe_2O_3 -Filled CNTs: CNTs with a diameter of $\approx 20 \text{ nm}$ were prepared by an AAO template method. The fabrication of the Fe_2O_3 -filled CNTs is described below.

Preparation of AAO Template: High-purity aluminum sheets were sonicated with acetone to degrease them and then etched in a sodium hydroxide (5.0 M) aqueous solution to remove impurities and the oxide layer on the surface. They were then anodized under a 20 V direct current for 2 h in a sulfur acid (20 wt%) aqueous solution at 10°C to obtain a nanoporous alumina layer on the surface. After removal of the as-grown alumina in a mixed acid of H_3PO_4 (6 wt%) and H_2CrO_4 (1.8 wt%) aqueous solution at 60°C , an ordered pore arrangement was

achieved after a second anodization of 4 h using the same conditions as for the first anodization. Free-standing AAO films with a pore size of $\approx 20 \text{ nm}$ were detached from the aluminum substrate by applying a voltage pulse, 5 V higher than the forming voltage in a perchloric acid-acetone (1:1, v/v) solution.

Preparation of Carbon-Coated AAO Films: The free-standing AAO films were first placed in a quartz boat which was inserted into a vertical tubular furnace. After heating to 650°C in an argon atmosphere at a rate of $10^\circ\text{C min}^{-1}$, acetylene was introduced into the furnace at $10 \text{ cm}^3 \text{ min}^{-1}$ with a $190 \text{ cm}^3 \text{ min}^{-1}$ flow of argon carrier gas for the chemical vapor deposition of carbon. The acetylene was shut off after 2 h deposition and the furnace was cooled to room temperature under the protection of an argon flow. As a result, carbon-coated AAO films were obtained.

Preparation of Fe_2O_3 Nanoparticle-Filled CNTs: Fe_2O_3 -filled CNTs were prepared by immersing the carbon-coated AAO films into a solution of iron nitrate in ethanol. First, carbon-coated AAO immersed in the ethanol solution was ultrasonicated for 3 h in an ice bath. Second, the film was heated to 80°C for 30 min and then rinsed with deionized water. Third, the material was heated to 140°C at 5°C min^{-1} under ambient conditions and held at this temperature for 6 h before being heated to 400°C at 2°C min^{-1} in an Ar atmosphere at which it was further held for 3 h. As a result, Fe_2O_3 nanoparticles from the decomposition of $\text{Fe}(\text{NO}_3)_3$ were loaded inside the nanochannels of the carbon-coated AAO film. Fe_2O_3 nanoparticle-filled CNTs were obtained by dissolving the AAO in a NaOH (5 M) aqueous solution.

Preparation of Fe_2O_3 Nanoparticle-Coated CNTs: 60 mg of CNTs prepared by the AAO template method was added to an 80 mL aqueous solution of $\text{Fe}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$ (15 wt%) to form a precursor suspension. After ultrasonication and stirring for 30 min, the CNTs were uniformly dispersed in the $\text{Fe}(\text{NO}_3)_3$ aqueous solution. The dispersion was then transferred to a 100 mL beaker placed inside a 250 mL beaker, and a 100 mL aqueous solution of ammonia (2 wt%) was added to the 250 mL beaker, which was then sealed by a film. NH_3 vapor then gradually diffused into the CNT dispersion, which was stirred magnetically to accelerate the hydrolysis of the $\text{Fe}(\text{NO}_3)_3$ formed. After continuous stirring for 4 h, the obtained suspension was removed from the sealed beaker and filtered. The collected product was washed with deionized water and dried at 80°C in air. The resulting material was heated to 140°C at 5°C min^{-1} under ambient conditions and held at this temperature for 6 h before heating to 400°C in an Ar atmosphere at 2°C min^{-1} and where it was held for 3 h. As a result, Fe_2O_3 nanoparticle-coated CNTs were obtained.

Material Characterization: The microstructures of the materials were characterized by TEM (JEOL JEM-2010 and FEI Tecnai F20) at 200 kV. The phase and crystallographic structure of the samples were characterized by XRD using a Rigaku D/max-2400 diffractometer equipped with a $\text{Cu K}\alpha$ radiation ($\lambda = 1.54056 \pm \text{\AA}$) over the angular range $10\text{--}85^\circ(2\theta)$. Laser Raman measurements were performed from 50 to 2000 cm^{-1} at room temperature using a Jobin Yvon LabRam HR800 micro Raman spectrometer with an excitation line of 632.8 nm having a laser spot size of about $1 \mu\text{m}$. The Fe_2O_3 contents were measured by thermogravimetric analysis using a Netzsch STA449C thermal analyzer in the temperature range $30\text{--}1000^\circ\text{C}$ at a heating rate of $10^\circ\text{C min}^{-1}$ under an air flow (50 mL min^{-1}).

Electrochemical Measurements: Active materials of Fe_2O_3 nanoparticle-filled CNTs and Fe_2O_3 nanoparticle-coated CNTs were mixed with conductive carbon black (Super P) and binder (polyvinylidene fluoride), with weight ratios of 70:15:15, in an *N*-methyl-2-pyrrolidone solvent to form an electrode slurry, which was then coated on a copper foil to form the working electrode. After solvent evaporation, the electrodes were dried in vacuum at 120°C for 12 h. To ensure good electrical contact, a pressure of 2 MPa was applied to the electrode after drying. 2032 stainless steel coin cells were used for electrochemical measurements. The coin cells were assembled inside an Ar-filled glovebox ($<1 \text{ ppm}$ of oxygen and water, Mbraun, Unilab). The electrolyte was 1 M LiPF_6 dissolved in a 1:1:1 (v/v/v) mixture of ethylene carbonate, propylene carbonate, and dimethyl carbonate (DMC). Lithium metal foil (1 mm thick) was used as

counter/reference electrodes and a Celgard 2400 membrane was used as separator. The cells were galvanostatically charged and discharged using a battery test system (LAND CT2001A model, Wuhan LAND Electronics Co., Ltd.) in the voltage range 0.01–2.8 V versus Li^+/Li at a current rate of 50 mA g^{-1} . The specific capacity with respect to the mass of Fe_2O_3 was calculated by the capacity of the electrode material after subtracting CNT capacity and then divided by the mass of Fe_2O_3 .

Prelithiation of the Fe_2O_3 -Filled CNTs and Fe_2O_3 -Coated CNTs: Free standing films consisting of Fe_2O_3 nanoparticle-filled CNTs or Fe_2O_3 nanoparticle-coated CNTs were prepared by a simple filtration method. First, both samples were dispersed in deionized water using sonication to form a suspension. The as-obtained suspension was then suction filtered. As a result, the suspension settled on the filter membrane and was collected as a thin sheet of self-supporting “paper” after drying. The dried CNT paper was cut into circular sheets (10 mm in diameter) and used as the working electrode materials for coin cells without any conductive agent and binder. The paper electrodes in the coin cells were discharged (lithiated) from the open circuit voltage to 0.01 V at a current density of 20 mA g^{-1} . The cells were disassembled in a glove box and the prelithiated samples of the working electrodes were removed, washed with anhydrous DMC, and dried in a vacuum for 2 h before use by inserting them in the transmission electron microscope.

Construction of the Prelithiated CNT-Based Nanobatteries: In situ TEM charge–discharge tests were conducted in an FEI Tecnai F20 TEM with a Nanofactory TEM-STM holder. The accelerating voltage was 200 kV. Before inserting the holder into the TEM, the prelithiated $\text{Fe}_2\text{O}_3/\text{CNT}$ samples were loaded onto the edge of a gold rod with a freshly cut tip to form an electrode by simply scratching the rod against a bulk powder of the as-obtained samples. A gold tip was mounted on a piezo-manipulator to form the counter electrode, which was driven to approach the prelithiated samples inside the TEM during the experiments. The lithiation was carried out at a positive bias in the range of + (4–5) V with respect to the counter electrode. The delithiation process was achieved using a negative bias in the range –(4–5) V. Note that both positive and negative bias voltages are needed because there is no reference electrode. By using different bias voltages in the discharge/charge processes, the reaction rate can be tuned.

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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